Essential in Creo (40 Hours)

S.No	Session	Topics	Type	Duration (Hrs)
1	Session 1	**Introduction:** Creo Parametric Basic Modeling Process **Understanding Creo Parametric Concepts:** Understanding Solid Modeling Concepts, Understanding Feature- Based Concepts, Understanding Parametric Concepts, Understanding Associative Concepts, Understanding Model-Centric Concepts, Recognizing File Extensions **Using the Creo Parametric Interface:** Understanding the Main Interface, Understanding the Folder Browser, Understanding the Ribbon Interface, Setting the Working Directory and Opening and Saving Files, Managing Files in Creo Parametric **Creating Sketcher Geometry:** Reviewing Sketcher Theory, Understanding Design Intent, Modifying the Sketcher Display, Utilizing Constraints, Sketching Lines, Centerlines, Rectangles, Circles, Arcs, Circular Fillets and Chamfers **Using Sketcher Tools**: Understanding Construction Geometry Theory, Sketching Points, Using Geometry Tools, Manipulating Sketches, Dimensioning Entities, Modifying Dimensions, Sketcher Conflicts **Creating Sketches for Features:** Creating Sketches ('Sketch' Feature), Specifying the Sketch Setup, Utilizing Sketch References, Using Entity from Edge within Sketcher	Theory/Demo	2 Hrs
2	Session 2	Exercises included for practice	Practicals	2 Hrs
3	Session 3	**Creating Extrudes, Revolves, and Ribs:** Creating Solid Extrude Features, Adding Taper to Extrude Feature, Common Dashboard Options: Extrude Depth, Common Dashboard Options: Feature Direction, Common Dashboard Options: Thicken Sketch, Creating Solid Revolve Features, Common Dashboard Options: Revolve Angle, Automatically Adding and Removing Material, Creating Rib Features. **Using the Creo Parametric Interface – Part 2:** Understanding Basic Display Options, Analyzing Basic 3-D Orientation, Understanding the View Manager, Creating and Managing View Orientations, Creating Style States using the View Manager, Understanding Basic Color and Appearance Options, Setting Up New Part Models.	Theory/Demo	2 Hrs
4	Session 4	Exercises included for practice	Practicals	2 Hrs
5	Session 5	**Selecting and Editing:** Understanding Creo Parametric Basic Controls, Using Drag Handles and Dimension Draggers, Using Keyboard Shortcuts, Model Tree, Selecting Items using Direct Selection and Query Selection, Using the Search Tool, Understanding Selection Filters, Renaming Objects, Activating and Editing Models, Deleting and Suppressing Items **Creating Datum Features: Planes and Axes:** Creating Datum Axes, Creating Datum Planes **Creating Sweeps and Blends:** Creating Sweeps with Open Trajectories, Creating Sweeps with Closed Trajectories, Analyzing Sweep Feature Attributes, Creating a Parallel Blend Protrusion or Cut, Experimenting with Parallel Blend Attributes, Analyzing Parallel Blend Section Tools	Theory/Demo	2 Hrs
6	Session 6	Exercises included for practice	Practicals	2 Hrs
7	Session 7	**Creating Holes and Shells:** Common Dashboard Options: Hole Depth, Creating Coaxial Holes, Creating Linear Holes, Creating Radial and Diameter Holes, Exploring Hole Profile Options, Creating Shell Features **Advanced hole Creation:** Creating Standard Holes, Creating Sketched Holes, Creating On Point Holes, Lightweight Holes, Cosmetic Threads **Creating Rounds and Chamfers:** Creating Rounds by Selecting Edges, by Selecting a Surface and Edge and by Selecting Two Surfaces, Creating Full Rounds, Creating Chamfers by Selecting Edges **Drafts:** Creating Draft Features, Creating Basic Split Drafts, Analyzing Draft Hinges and Pull Direction, Drafting Intent Surfaces, Creating Drafts with Multiple Angles, Using the Extend Intersect Surfaces Draft Option, Creating Drafts Split at Sketch, Creating Drafts Split at Curve, Creating Drafts Split at Surface, Creating Drafts with Variable Pull Direction	Theory/Demo	2 Hrs
8	Session 8	Exercises included for practice	Practicals	2 Hrs
9	Session 9	**Variable Section Sweeps:** Creating Variable Section Sweeps using a Constant Section, Creating Variable Section Sweeps Normal to Trajectory, Creating Variable Section Sweeps using Constant Normal Direction, Creating Variable Section Sweeps Normal to Projection, Creating Variable Section Sweeps Utilizing Multiple Trajectories. **Helical Sweeps:** Understanding Helical Sweep Theory, Creating Helical Sweeps for Springs, Creating Helical Sweeps for Threads, Analyzing Helical Sweep Profile and Pitch Variations, Utilizing Variable Sections in Helical Sweeps	Theory/Demo	2 Hrs

		Swept Blends: Understanding Swept Blend Theory, Creating Swept Blends by Selecting Sections, Creating Swept Blends by Sketching Sections, Analyzing Swept Blend Section Options, Analyzing Swept Blend Section Plane Control, Analyzing Swept Blend Options.		
10	Session 10	Exercises included for practice	Practicals	2 Hrs
11	Session 11	**Creating Patterns:** Direction Patterning in the First Direction, Direction Patterning in the Second Direction, Axis Patterning in the First Direction, Axis Patterning in the Second Direction, Creating Reference Patterns of Features, Creating Reference Patterns of Components **Group, Copy, and Mirror Tools:** Creating Local Groups, Copying and Pasting Features, Moving and Rotating Copied Features, Mirroring Selected Features, Mirroring All Features, Creating Mirrored Parts **Measuring and Inspecting Models:** Analyzing Model Units and Mass Properties, Measuring Models, Creating Planar Part Cross-Sections, Measuring Global Interference **Relations and Parameters:** Understanding Relation Theory, Understanding Relation Types, Understanding Basic Relation Operators and Functions, Understanding Advanced Relation Operators and Functions, Creating Parameters, Understanding Advanced Parameter Options, Creating Relations, Creating Relations for Patterns **Family Tables:** Understanding Family Table Theory, Creating a Family Table, Patterning Family Table Instances, Creating a Multi-Level Family Table, Editing Family Table Members	Theory/Demo	2 Hrs
12	Session 12	Exercises included for practice	Practicals	2 Hrs
13	Session 13	**Assembling with Constraints:** Understanding Assembly Theory, Creating New Assembly Models, Understanding Constraint Theory, Understanding Assembly Constraint Status, Assembling Components using: Default, Automatic, Distance, Angle Offset, Parallel, Coincident, Normal, Coplanar, Centered, Tangent, Fix Constraints **Exploding Assemblies:** Creating and Managing Explode States, Creating Offset Lines between Exploded Components	Theory/Demo	2 Hrs
14	Session 14	Exercises included for practice	Practicals	2 Hrs
15	Session 15	**Surface Features:** Creating Surface features, Editing Surface Features in Creo Parametric, Copying the Surfaces, Mirroring the Surfaces, Trimming the Surface, Merging the Surfaces, Creating the Fill Surface, Creating the Intersecting Curves, Creating an Offset Surface, Giving Thickness to a Surface, Converting a Surface into a Solid **Creating Drawing Views:** Analyzing Drawing Concepts and Theory, Analyzing Basic 2-D Orientation, Creating New Drawings and Applying Formats, Creating and Orienting General Views, Adding Drawing Models and Sheets, Creating Projection Views **Creating Drawing Views:** Creating Cross-Section Views, Creating Detailed Views, Creating Auxiliary Views, Creating 2D Cross Section Views, Creating New Drawings using Drawing Templates, Modifying Drawing Views, Creating Assembly and Exploded Views **Creating Drawing Details:** Analyzing Detail Concepts and Types, Showing and Erasing Detail Items, Cleaning Up Dimensions, Manipulating Dimensions, Creating Driven Dimensions **Creating Drawing Details:** Creating Notes, Showing a Bill of Materials, Analyzing Drawing Associativity	Theory/Demo	2 Hrs
16	Session 16	Exercises included for practice	Practicals	2 Hrs
17	Session 17	**Creating and Using Component Interfaces:** Understanding Component Interfaces, Using a Placing Component Interface, Using a Receiving Component Interface, Creating a Component Interface using Save as Interface, Auto Placing Components **Replacing Components in an Assembly:** Understanding Component Replace, Replacing Components using Family Table **Understanding the Basics of Simplified Reps:** Understanding Simplified Rep Concepts, Customizing Simplified Reps, Excluding Components using Simplified Reps, Creating Geometry and Graphics Simplified Reps, Opening Simplified Reps, Creating Part Simplified Reps	Theory/Demo	2 Hrs
18	Session 18	Exercises included for practice	Practicals	2 Hrs
		Introduction to sheet metal design: The sheet metal design environment, Sheet metal parts, Design approach **Primary walls:** Wall Types, Creating the first wall **Secondary and unattached walls:** Types of secondary walls, Creating unattached walls, Secondary walls with relief		

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19	Session 19	**Bend features:** Bend features, Bend lines **Unbend, bend back and cuts:** The unbend regular feature, The bend back feature, Creating sheet metal cuts **Notches and punches:** Creating a punch or notch UDF, Placing a punch or notch feature **Sheet metal forms:** Creating form features, Creating form parts, Placing form features, Returning the model to the flat state	Theory/Demo	2 Hrs
20	Session 20	Exercises included for practice	Practicals	2 Hrs